A Comparative Analysis of Existing Research Relevant to Tatkal Railway Ticket Booking

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1. Introduction

Tatkal railway ticket booking is a highly time-sensitive process in which a large number of passengers attempt to secure a limited number of seats within a short booking window. During peak booking hours, users may experience difficulties such as frequent logouts, slow website responses, transaction failures, CAPTCHA reloads, and the absence of suitable alternative trains when their preferred option is unavailable. These issues can make the booking process uncertain and frustrating, even when alternative travel options may exist.

Recent research has explored different aspects of intelligent railway and transportation systems, including passenger demand estimation, route recommendation, personalized travel choices, Tatkal confirmation prediction, and train-selection decision support. This analysis examines five such studies published between 2024 and 2026.

This work presents a focused comparative analysis of five existing studies related to railway demand estimation, route recommendation, passenger personalization, Tatkal confirmation prediction, and train-selection decision support. The objective is to identify common approaches, limitations, and research gaps relevant to the development of an intelligent Tatkal booking system.

2. Overview of the Selected Studies

Paper	Year	Problem Addressed	Method	Dataset	Main Contribution
[1] Multilayered Railway Passenger Demand Estimation Considering Nested Choices: A Computational Graph-Based	2025	Estimating passenger demand at multiple railway levels while considering passenger choices and capacity restrictions	Computational graph-based learning framework with passenger-choice modelling and capacity considerations	Beijing–Shanghai high-speed railway corridor	Integrates passenger behaviour, demand estimation and railway capacity into a unified framework

Learning Framework					
[2] Travel Route Recommendation with a Trajectory Learning Model	2024	Recommending suitable travel routes based on historical travel behaviour and contextual conditions	Trajectory learning, graph neural networks and sequential learning	Four real-world travel trajectory datasets	Uses historical trajectories, road-network information, traffic and weather to generate route recommendations
[3] Personalized Route Recommendation for Passengers in Urban Rail Transit Based on Collaborative Filtering Algorithm	2024	Providing personalized route recommendations according to passenger preferences and congestion	Collaborative filtering and congestion-aware recommendation	Shenzhen Metro	Uses individual and similar-user travel behaviour along with travel time, transfers and congestion
[4] Tatkal Ticket Booking Probability Prediction System: A Route-Aware Intelligent Railway Advisory Approach	2026	Predicting the probability of successful Tatkal ticket confirmation	Logistic Regression, Bayesian Smoothing and Weighted Linear Ensemble	Synthetic dataset containing 66,636 records	Provides route/train-specific Tatkal confirmation probabilities and a practical advisory interface

[5] On a Hybrid Decision Support Framework for Train Selection in Indian Railways	2026	Supporting train selection under uncertainty in ticket availability	DRSA/CRSA, XGBoost and SVM	Indian Railways/IRCTC reservation context	Combines interpretable decision rules with machine-learning-based train-rating prediction

3. Thematic Analysis

3.1 Evolution of Methodological Approaches

The five studies demonstrate different levels of intelligent decision-making in railway and transportation systems.

Study [1] focuses on the system-level perspective by estimating passenger demand at multiple levels while considering capacity restrictions. This is relevant to Tatkal booking because understanding historical passenger demand can help estimate how competitive a particular train or route may become.

Study [2] focuses on intelligent route recommendation. Its trajectory learning approach uses historical travel trajectories together with road-network structure and contextual factors such as traffic and weather to generate route recommendations. This demonstrates how historical behaviour can be transformed into practical travel suggestions.

Study [3] places greater emphasis on personalization. Through collaborative filtering, the system considers individual preferences and the behaviour of similar passengers when recommending routes. Travel time, transfers and congestion are incorporated into the recommendation process.

Study [4] directly addresses the Tatkal problem by predicting ticket confirmation probability. It combines Logistic Regression, Bayesian Smoothing and a Weighted Linear Ensemble. The reported ensemble achieved 74.10% accuracy and an AUC of 0.79. However, the evaluation was conducted on synthetic data and used in-sample testing,

which limits the extent to which these results can be generalized to actual Tatkal booking conditions [4].

Study [5] adopts a decision-support perspective by combining rough-set-based decision rules with machine-learning models for train selection. This approach is relevant when passengers need to compare several train options using multiple criteria.

3.2 Common Findings Across the Studies

A major similarity among the five studies is the importance of historical data for understanding travel behaviour and generating predictions or recommendations [1]–[5].

Study [1] uses demand-related information to understand passenger flows and choices, while [2] learns travel behaviour from historical trajectories. Study [3] uses previous passenger behaviour to identify preferences, and [4] uses synthetic booking-related data to estimate Tatkal confirmation probability. Study [5] similarly uses preference and historical information to support train selection.

Another common theme is that travel decisions are influenced by multiple factors rather than a single variable. Depending on the study, these factors include passenger demand, route characteristics, travel time, congestion, transfers, timing, distance, preferences and ticket availability [1]–[5].

This is particularly relevant to Tatkal booking. A passenger may prefer a particular train, but if the expected confirmation probability is low, another train with slightly different characteristics may provide a better overall option.

3.3 Differences Between the Studies

The primary difference between the studies lies in what each system attempts to predict or optimize.

Study [1] concentrates on aggregate railway passenger demand and capacity, making it more system-oriented. Studies [2] and [3] concentrate on route recommendation, with [3] adding a stronger personalization component.

Study [4] specifically predicts Tatkal ticket confirmation probability, making it the most directly relevant study to the proposed project. Study [5], in contrast, focuses on train selection and decision support, where multiple criteria are used to determine suitable options.

Therefore, the studies are not necessarily contradictory. Instead, they represent different but complementary objectives within the larger railway travel problem.

3.4 Real-Time Decision-Making

Real-time applicability is an important distinction across the literature.

Several of the reviewed approaches depend heavily on historical information. Historical patterns are useful for prediction, but they may not completely represent the conditions of a particular Tatkal booking window.

Study [3], for example, considers congestion and personalization but does not address real-time ticket or seat availability, booking failures, or Tatkal confirmation probability [3].

Study [4] provides a practical interface for obtaining route- and train-specific Tatkal confirmation estimates, but its underlying evaluation is based on a synthetic dataset rather than actual historical IRCTC booking records [4].

Study [5] addresses uncertainty in train selection, but the available study information does not establish the extent of its real-time seat-availability validation [5].

This highlights an important distinction between predicting booking conditions and supporting a passenger during an actual high-demand booking process.

4. Strengths and Limitations Across the Studies

4.1 Strengths

The five studies collectively provide several useful foundations for an intelligent Tatkal booking system.

Study [1] contributes the concept of demand estimation and capacity-aware modelling, which can help identify trains or routes likely to experience high demand.

Study [2] demonstrates how historical travel behaviour and contextual information can be used to generate alternative routes.

Study [3] introduces personalized recommendation, showing that travel suggestions can be adapted to individual preferences and similar-user behaviour.

Study [4] provides the most direct foundation for Tatkal booking

Study [5] contributes the idea of combining multiple decision criteria with machine-learning predictions, which is useful when selecting the most suitable alternative train.

Together, the studies suggest that an effective railway booking assistant could benefit from combining prediction, recommendation, personalization and decision support.

4.2 Limitations

Despite these contributions, several limitations are visible across the reviewed studies.

First, many of the approaches rely heavily on historical data [1]–[5]. While historical data can identify useful patterns, actual booking conditions can change rapidly, particularly during high-demand periods.

Second, the studies generally focus on individual components of the travel decision rather than the complete booking workflow. A route recommendation system does not necessarily indicate whether a ticket will be successfully booked, while a confirmation prediction system does not necessarily account for individual passenger preferences.

Study [4] has a particularly important limitation for the proposed Tatkal problem. Its dataset consists of 66,636 synthetic records, and its reported evaluation is in-sample. Consequently, the reported 74.10% accuracy and 0.79 AUC demonstrate the behaviour of the proposed model on the experimental data but do not establish its performance on real-world IRCTC booking conditions [4].

Furthermore, the selected studies do not collectively address practical booking difficulties such as session loss, transaction failure, CAPTCHA reloading and peak-hour system slowdown as part of one integrated system.

5. Research Gap

The comparison reveals a gap between individual intelligent railway functions and an integrated Tatkal booking assistance system.

Existing research provides approaches for:

- estimating passenger demand [1],
- learning travel and route behaviour [2],
- recommending personalized routes [3],
- predicting Tatkal confirmation probability [4], and
- supporting train selection using multiple criteria [5].

However, these capabilities remain largely separated across different studies.

Based on the reviewed literature, there is an opportunity to develop a Tatkal-focused system that combines:

Tatkal probability prediction + alternative train/route recommendation + passenger preferences + transaction preparation + failure-aware decision support.

Another important gap concerns the broader booking experience during peak hours. The selected studies primarily concentrate on prediction, recommendation or decision support rather than simulating how the system can respond when the preferred booking option becomes unavailable or a transaction attempt fails.

The proposed project therefore investigates whether these separate intelligent functions can be combined within a single web-based Tatkal booking simulation.

6. Relevance to the Proposed Work

The proposed project is a web-based simulation of an intelligent Tatkal railway ticket booking system. Its primary objective is to explore how AI-based prediction and decision-support mechanisms can improve the reliability and flexibility of the booking process during high-demand periods.

The approaches identified in the five studies can be mapped to different components of the proposed system.

6.1 AI-Based Seat Availability Prediction

The demand-estimation concept in [1] and the Tatkal probability prediction approach in [4] provide a foundation for developing an AI model that learns from past booking or traffic trends.

The proposed simulation can use dummy historical data containing factors such as route, train, journey date, demand and availability to estimate the probability of obtaining a Tatkal ticket.

Instead of providing only a binary result such as “available” or “unavailable,” the system can produce a probability or confidence estimate that helps the user understand the expected difficulty of booking.

6.2 Intelligent Tatkal Booking Suggestions

The proposed system can use its predicted probability to provide an early booking recommendation.

For example, if the predicted probability for a preferred train is low, the system can alert the user and recommend preparing an alternative option before the Tatkal booking window.

The project also proposes a simulated transaction-preparation mechanism, in which payment-related information or transaction preparation is completed before the booking window. In the simulation, this can represent an attempt to reduce unnecessary processing during the peak booking period.

Because the proposed project is a simulation, this functionality should be described as a prototype or simulation of the concept, rather than a claim that it can directly eliminate payment failures on the actual railway reservation system.

6.3 Alternative Train and Route Recommendation

The route-recommendation approaches presented in [2] and [3] are particularly relevant to the project's backup-option mechanism.

If the preferred train or route has a very low predicted Tatkal confirmation probability, the system can search for alternative trains or routes.

These alternatives can be ranked according to factors such as:

- travel distance,
- travel time,
- number of train changes,
- preferred stations,
- predicted Tatkal probability, and
- user-defined preferences.

This transforms the system from a simple prediction model into a decision-support system.

6.4 Preference-Based Train Selection

The personalization concept from [3] can be incorporated through a user preference profile.

For example, a user may prioritize:

shorter travel time → fewer train changes → preferred station → higher confirmation probability

while another user may prioritize:

higher confirmation probability → fewer changes → shorter distance.

The proposed system can use these preferences to rank alternative trains differently for different users.

This is an important extension because the train with the highest confirmation probability may not always be the most suitable train for a particular passenger.

6.5 Failure-Aware Booking Simulation

A distinctive aspect of the proposed project is that it considers what happens after the preferred booking option becomes unsuccessful.

Instead of treating an unsuccessful booking attempt as the end of the process, the system can simulate a backup decision:

Preferred Train → Booking Attempt → Failure / Low Probability → Evaluate Alternatives → Select Next-Best Train → Booking Simulation

This connects the prediction, recommendation and decision-support components into one workflow.

The multi-criteria decision-support perspective discussed in [5] is relevant here because the system must evaluate several factors before selecting the next-best train.

Research	Contribution	Our System
[1]	Demand estimation	Availability prediction
[2]	Route recommendation	Alternative routes
[3]	Personalization	User preferences
[4]	Tatkal probability	Tatkal prediction

[5]	Decision support	Backup selection

7. Synthesis

The five studies collectively show a progression in intelligent transportation research from demand estimation and behavioural modelling toward personalized recommendation, prediction and decision support.

Study [1] demonstrates how passenger demand and capacity can be modelled. Studies [2] and [3] show how route recommendations can incorporate historical behaviour and personalization. Study [4] provides a direct approach for predicting Tatkal confirmation probability, while [5] demonstrates how multiple criteria and machine-learning techniques can support train selection.

The proposed project brings these ideas together around the Tatkal booking scenario.

Rather than functioning only as a prediction tool, the proposed system aims to simulate a complete decision-support workflow in which the system can:

1. predict Tatkal availability or confirmation probability;
2. identify when the preferred option has a low probability of success;
3. recommend alternative trains or routes;
4. rank alternatives according to user preferences;
5. simulate transaction preparation before the booking window; and
6. provide a backup option when the primary booking attempt fails.

The proposed work should therefore be positioned as an integrated proof-of-concept and web simulation, rather than as a fully validated replacement for the existing railway reservation system.

8. Potential Contribution of the Proposed Work

The primary contribution of the proposed project lies in integrating several existing research directions into one Tatkal-oriented workflow.

The individual techniques are not necessarily novel by themselves. Demand prediction, route recommendation, collaborative filtering, probability prediction and decision-support methods have already been explored in the reviewed literature [1]–[5].

The potential contribution instead lies in their combination and application to the Tatkal booking scenario.

The proposed system attempts to shift the focus from:

> “Can I book this particular train?”

to:

> “Considering my preferences and the predicted booking conditions, what is my best available option and what should I do if my first choice fails?”

This decision-oriented approach provides a broader perspective on the Tatkal booking problem and creates a foundation for future research using real historical booking data and real-time railway information.

9. Conclusion

The comparative analysis of five studies demonstrates that existing research has addressed several important components of intelligent railway and travel systems, including passenger demand estimation [1], route recommendation [2], personalized travel decisions [3], Tatkal confirmation prediction [4], and multi-criteria train-selection decision support [5].

However, these capabilities remain largely distributed across separate systems. In particular, the reviewed literature does not provide, based on the available information, a single Tatkal-focused framework that integrates probability prediction, alternative train and route recommendation, passenger preferences, transaction preparation and failure-aware decision support.

The proposed web-based simulation attempts to address this gap by combining these components into a unified Tatkal booking assistance workflow. Since the project primarily uses dummy data, its results should be interpreted as a proof of concept rather than evidence of performance on the actual railway reservation infrastructure.

Future work could involve evaluating the proposed approach using larger real historical booking datasets, incorporating real-time availability information, improving prediction models, and testing whether intelligent alternative selection can improve the efficiency and user experience of Tatkal booking under realistic peak-hour conditions.

References

[1] X. Wang, H. Fu, X. Wu, Y. Liu, and T. Kim, “Multilayered Railway Passenger Demand Estimation Considering Nested Choices: A Computational Graph-Based Learning Framework,” *Urban Rail Transit*, 2025.

[2] X. Wu, Z. Zhang, and W. Wan, “Travel Route Recommendation with a Trajectory Learning Model,” *Complex & Intelligent Systems*, 2024.

[3] W. Li, Z. Li, and Q. Luo, “Personalized Route Recommendation for Passengers in Urban Rail Transit Based on Collaborative Filtering Algorithm,” *IET Intelligent Transport Systems*, 2024.

[4] R. Panchal, G. Patel, S. Bole, A. Dubey, and I. Mahto, “Tatkal Ticket Booking Probability Prediction System: A Route-Aware Intelligent Railway Advisory Approach,” *International Journal of Engineering Development and Research (IJEDR)*, vol. 14, no. 2, 2026.

[5] “On a Hybrid Decision Support Framework for Train Selection in Indian Railways: An Integration of Dominance-Based Rough Set Approaches and Machine Learning Models,” *Transport and Telecommunication Journal*, vol. 27, no. 1, 2026.